

Bluestock MF Capstone Project

Final Report

Prepared by: Anil Kankala

B.Tech, Artificial Intelligence and Data Science

VNR VJIET, Hyderabad

1. Executive Summary

This project builds an end-to-end data pipeline for analyzing Indian mutual fund data, covering 40 schemes across 10 fund houses from 2022 to 2025. The pipeline ingests raw CSV datasets, cleans and validates them, loads the results into a SQLite star schema, performs exploratory data analysis, and presents findings through an interactive Power BI dashboard.

Industry AUM grew from roughly Rs 3M Cr in 2022 to over Rs 6M Cr by 2025, with SBI Mutual Fund holding the largest share among all 10 fund houses tracked, followed by ICICI Prudential and HDFC. Lumpsum investments account for the largest share of transaction value (58.5%), followed by Redemptions (35.3%), with SIP contributing the smallest share (6.2%). Investors aged 26-35 invest the highest average SIP amount, with contribution declining across older age groups.

2. Data Sources

The project uses 10 raw CSV datasets, plus live NAV data fetched directly from the `mfapi.in` public API:

- `01_fund_master.csv` - one row per fund (40 funds, 10 fund houses, categories, sub-categories, risk levels)
- `02_nav_history.csv` - daily NAV per fund, 2022-2025
- `03_aum_by_fund_house.csv` - AUM tracked per fund house over time
- `04_monthly_sip_inflows.csv` - industry-wide SIP inflow trends
- `05_category_inflows.csv` - net inflows by fund category
- `06_industry_folio_count.csv` - investor folio counts over time
- `07_scheme_performance.csv` - returns and risk metrics per fund
- `08_investor_transactions.csv` - individual investor transactions
- `09_portfolio_holdings.csv` - stock-level holdings per fund
- `10_benchmark_indices.csv` - daily closing values for benchmark indices (NIFTY50, etc.)

Full column-level documentation is available in `data_dictionary.md` in the project repository.

3. ETL Design

3.1 Ingestion

All 10 raw datasets were loaded with pandas and inspected for shape, data types, and anomalies (missing values, duplicate rows, date columns stored as text, duplicate codes). Live NAV data for 6 schemes was additionally fetched from the mfapi.in API and saved as raw CSVs.

3.2 Cleaning

- `nav_history`: dates parsed to datetime, sorted by fund and date, missing NAV values forward-filled for weekends/holidays, duplicates removed, invalid NAV (≤ 0) dropped
- `investor_transactions`: `transaction_type` values standardised to SIP/Lumpsum/Redemption, invalid amounts (≤ 0) dropped, date format converted from DD-MM-YYYY to a proper date type
- `scheme_performance`: return values validated as numeric, expense ratio checked against the expected 0.1%-2.5% range

3.3 Star Schema

Cleaned data was loaded into a SQLite database (`bluestock_mf.db`) using a star schema with two dimension tables and four fact tables:

- `dim_fund` - one row per fund, primary key `amfi_code`
- `dim_date` - calendar dimension built from every date seen across the fact tables
- `fact_nav` - daily NAV per fund
- `fact_transactions` - every investor transaction
- `fact_performance` - one row per fund with performance and risk metrics
- `fact_aum` - AUM by fund house over time

Full CREATE TABLE statements are in `schema.sql`; 10 analytical queries are in `queries.sql`.

4. EDA Findings

Exploratory analysis was carried out in a Jupyter notebook (`EDA_Analysis.ipynb`), producing 10 charts covering NAV trends, AUM growth, SIP inflows, category inflows, investor demographics, geographic distribution, folio growth, return correlation, and sector allocation.

- NAV & AUM trend: industry AUM grew steadily from roughly Rs 3M Cr (2022) to over Rs 6M Cr (2025), with growth accelerating from 2024 onward
- AUM by AMC: SBI Mutual Fund leads all 10 fund houses, followed by ICICI Prudential and HDFC; DSP and Mirae Asset hold the smallest shares
- Investor demographics: the 26-35 age group contributes the highest average SIP amount, declining steadily through older age brackets
- Geography: Punjab, Tamil Nadu, and Madhya Pradesh lead in total transaction amount by state, indicating meaningful regional concentration
- Transaction mix: Lumpsum investments dominate transaction value (58.5%), followed by Redemptions (35.3%); SIP is the smallest share by value (6.2%)
- [Add: sector allocation and correlation matrix findings from your notebook - these aren't covered by the dashboard screenshots above]

5. Performance Analysis

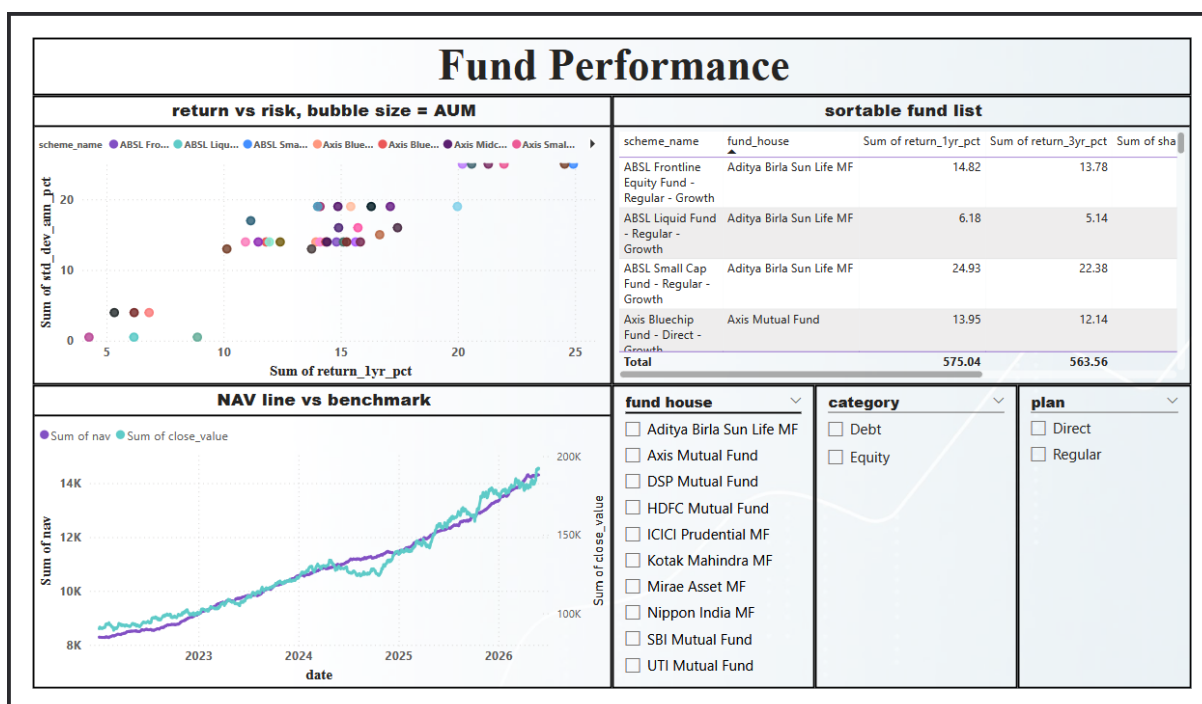
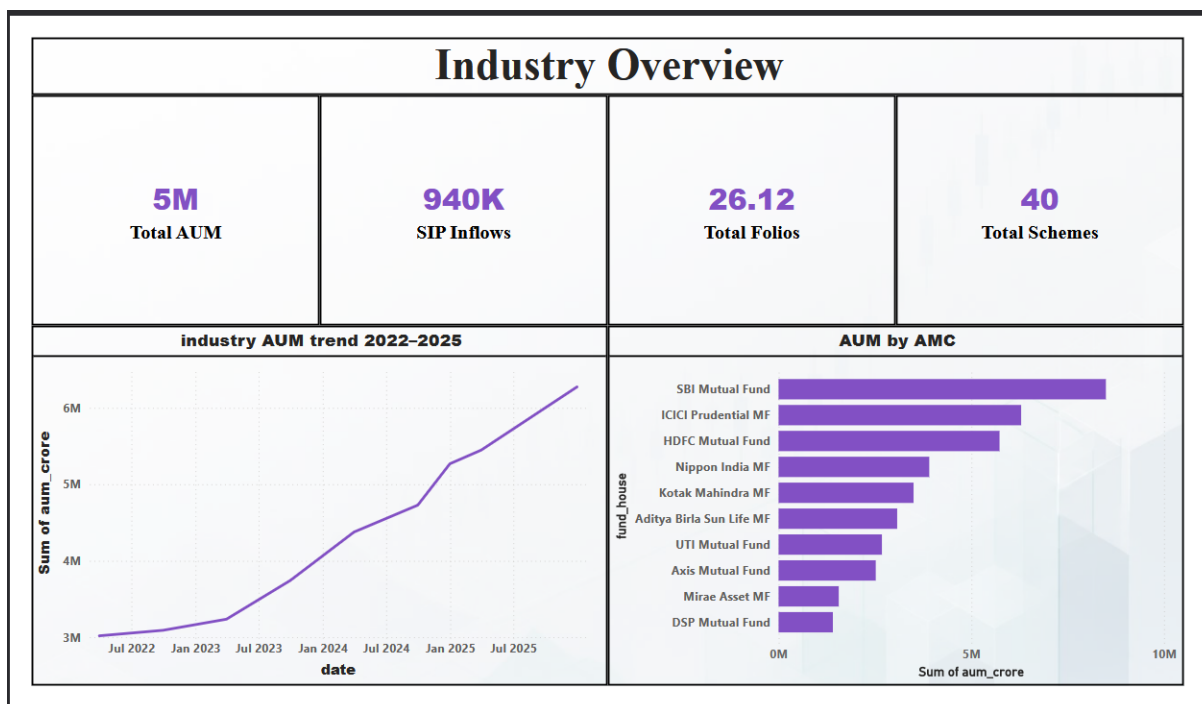
Fund performance was assessed using 1-year, 3-year, and 5-year returns, alongside risk-adjusted metrics (Sharpe ratio, Sortino ratio, standard deviation, max drawdown) and expense ratios. A pairwise correlation matrix of daily NAV returns was computed across 10 selected funds to identify diversification patterns.

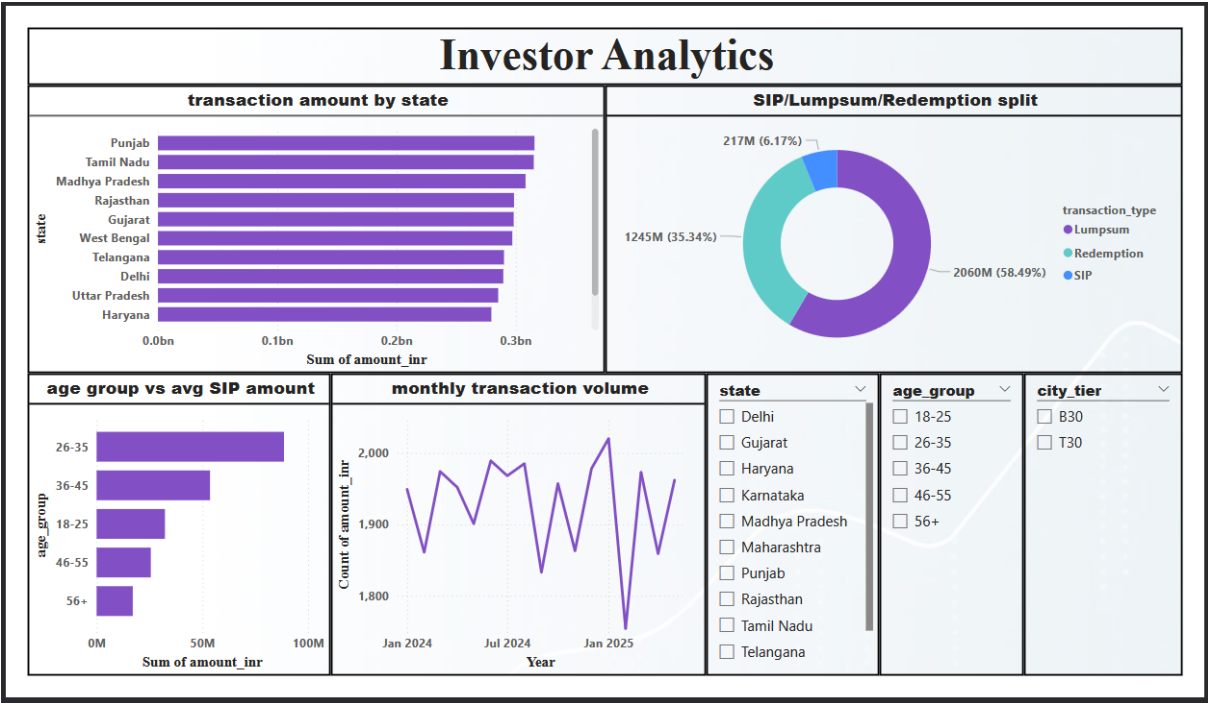
The fund scorecard (dashboard Page 2) allows sorting by return, Sharpe ratio, and expense ratio to identify top performers on a risk-adjusted basis.

[Add: name the top 2-3 funds by Sharpe ratio from your scorecard, and note any correlation patterns from your notebook's heatmap - e.g. which fund pairs move closely together vs independently, which matters for portfolio diversification.]

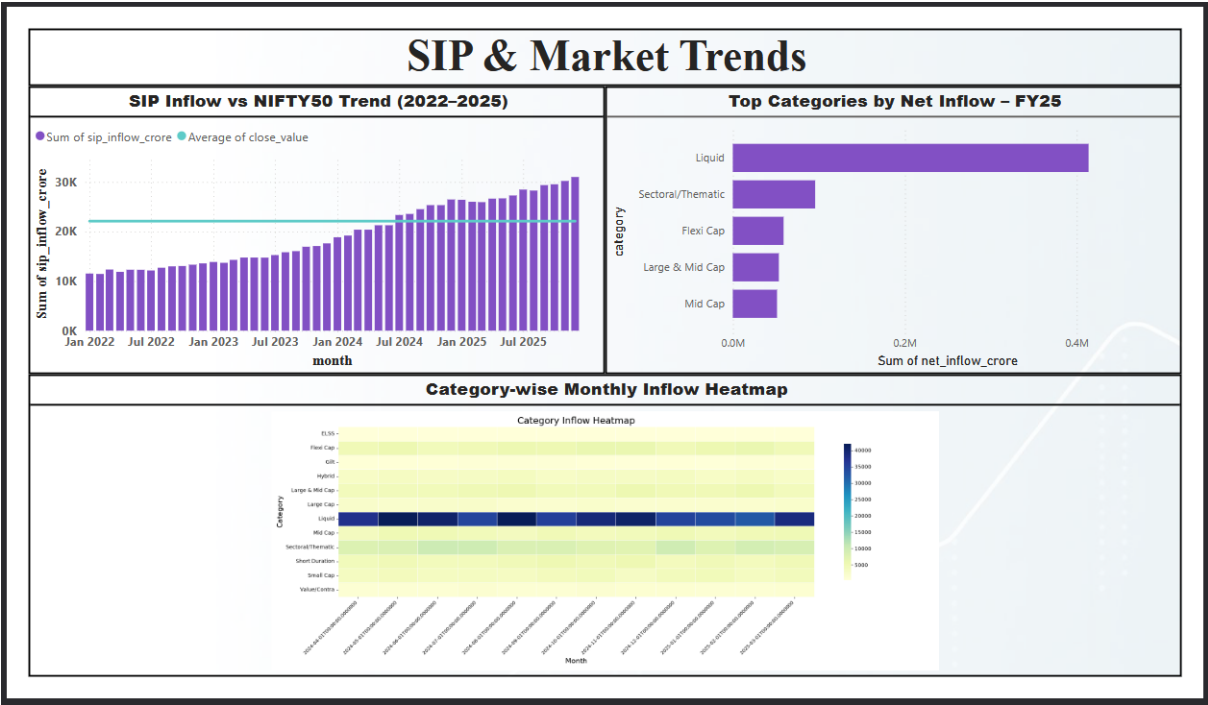
6. Dashboard Screenshots

The Power BI dashboard (bluestock_mf_dashboard.pbix) has 4 pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.





Page 3: Investor Analytics - transactions by state, SIP/Lumpsum/Redemption split, demographics



7. Limitations

- Dataset is limited to 40 schemes across 10 fund houses - not representative of the full Indian mutual fund industry (which has 1,900+ schemes)
- NAV gaps on weekends/holidays were forward-filled rather than using actual published values, which is a simplification
- Live NAV fetch depends on the availability and uptime of the mfapi.in public API
- Investor transaction data is limited to the fields provided and may not capture the full diversity of real-world investor behavior

8. Recommendations

- Expand the dataset to cover more schemes and fund houses for a more representative analysis
- Automate the pipeline on a schedule (e.g. daily NAV refresh) rather than running it manually
- Add data validation alerts so anomalies are flagged automatically rather than requiring manual review
- Extend the dashboard with drill-through pages and more granular investor segmentation